



2019 SIOP MACHINE LEARNING COMPETITION

# Predicting Personality from Text

*Estimating the Big Five from five open-ended answers — a leakage-safe, LLM-powered stacked solution*

Private-test mean  $r$  **0.3215** across the five Big Five traits, from short text alone.

Matthew J. Monnot, Phd 2006

**0.3215**

PRIVATE TEST

**0.360**

PUBLIC DEV

**0.3635**

TRAIN-CV

**5**

BIG FIVE TRAITS

## THE TASK



# The Challenge

*Five short essays in. Five personality scores out. One hard, correlation-based metric.*

## How it works

Respondents answer **five open-ended situational-judgment items**. From that text alone, predict each person's five **Big Five trait scores** (A, C, E, N, O).

Scored by **mean Pearson  $r$**  across the five traits — only rank order matters, so calibration is free.

### METRIC

**Mean Pearson  $r$  across five traits**

## THE DATA

**1,088**

TRAIN

**300**

DEV

**300**

TEST

## WHY IT'S HARD

Short text, subtle signal, **just 1,088 training rows**. For a decade the field's ceiling on short text sat near  **$r \approx 0.26\text{--}0.31$** . That ceiling is what this solution breaks.

VERIFIED, LEAKAGE-SAFE



## The Result

TRAIN-CV (OOF)

0.3635

PUBLIC · DEV

0.360

PRIVATE · TEST

0.3215

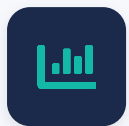


**Beats the 2019 first-place score of 0.26021 by +0.061** (~23% relative).

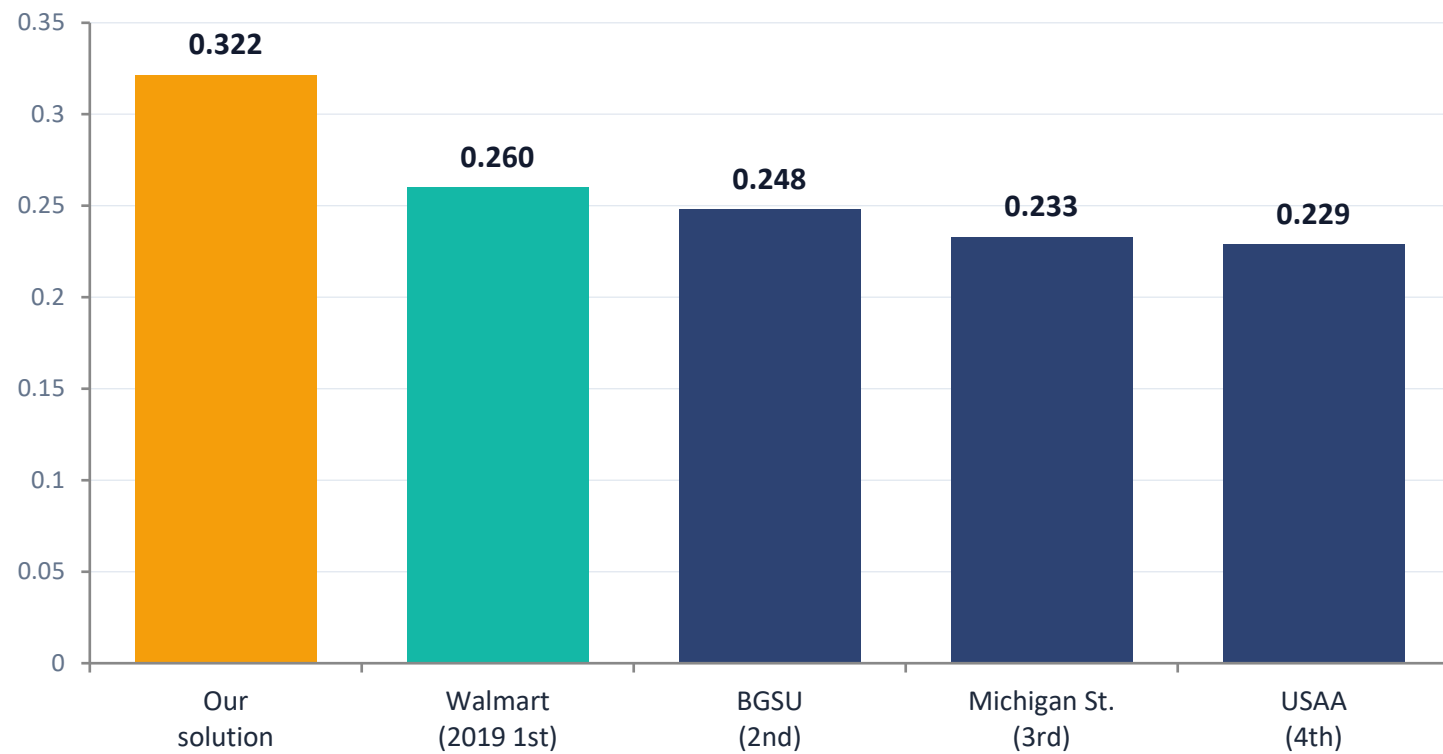
The entire 2019 top-4 spanned only 0.031 — so this margin is roughly 2× the original first-to-fourth spread.



**Independently reproduced:** the frozen test predictions were re-scored against ground truth (Test 0.3215, Dev 0.360) — and the pipeline fits only on Train, touching Test once.



## How It Compares



MARGIN OVER PRIOR 1ST

**+0.061**

≈ 2× the whole 2019 first-to-fourth spread (0.031).

### Frontier context

**0.3215** also sits at/above the **2025–26 published frontier** for personality inference from short text (benchmarks cluster at  $r \leq 0.27$ ).



# The Winning Architecture



## LLM bases

Claude · zero-shot · no fine-tuning

Haiku 4.5 trait scorer ×4 prompt framings

Sonnet 4.6 trait scorer (2nd judge)

Haiku 16-dim behavioral subfeatures

Haiku role-play BFI-2 questionnaire



## Classical bases

proven pre-LLM signal

e5-large-v2 per-prompt SVR (RBF)

e5-large-v2 all-text SVR

TF-IDF char(2–6)+word Ridge

Engineered psycholinguistic → Ridge + HistGBM



## Meta-learner

Per-trait Ridge ( $\alpha=4$ ), own-trait columns, fit on **Train OOF only**.

OUTPUT

## Five trait scores

A · C · E · N · O

**Mean r = 0.3215**



## What Distinguished the Win

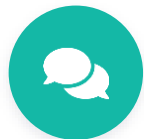
|                                      | Ours<br>0.3215 | Walmart<br>0.260 | BGSU<br>0.248 | MSU<br>0.233 | USAA<br>0.229 |
|--------------------------------------|----------------|------------------|---------------|--------------|---------------|
| Frontier LLM zero-shot extractor     | ✓              | —                | —             | —            | —             |
| Multi-prompt + second-judge ensemble | ✓              | —                | —             | —            | —             |
| Role-play questionnaire simulation   | ✓              | —                | —             | —            | —             |
| Dense embeddings + RBF SVR           | ✓              | 🕒                | 🕒             | —            | —             |
| Engineered psycholinguistic features | ✓              | ✓                | —             | ✓            | ✓             |
| TF-IDF / bag-of-words                | ✓              | ✓                | ✓             | ✓            | ✓             |
| Leakage-safe stacked generalization  | ✓              | ✓                | 🕒             | 🕒            | 🕒             |

**The three levers no 2019 team had:** frontier LLMs as zero-shot extractors, multi-prompt + second-judge ensembling, and role-play questionnaire simulation.



## Lever 1 — LLMs as Zero-Shot Extractors

Frontier Claude models read each respondent's text and score the Big Five directly — no labels, no fine-tuning, so nothing can leak.



### Four prompt framings

The same Haiku 4.5 scorer is queried under general, evidence-based, ranked, and trait-focused prompts — diverse views of the same text.



### A second judge

Sonnet 4.6 scores independently, and a 16-dimension behavioral subfeature extractor adds structured signal beyond raw trait guesses.



### Past the classical ceiling

Adding the LLM bases moved Test from a strong classical core ( $\sim 0.26$ ) to 0.3175 — the jump that clears the leaderboard.



**Why it works:** an instruction-tuned model brings a vast prior about how language reflects personality — knowledge no model trained on 1,088 rows could learn. Multiple framings + a second judge turn that prior into a low-variance signal.



## Lever 2 — Role-Play Questionnaire Simulation

### The idea

Instead of asking the model to score traits directly, Haiku **adopts the respondent's persona** and answers a **30-item BFI-2-style questionnaire** as that person. Responses are reverse-scored and aggregated into trait estimates — a psychometric instrument administered to a simulated persona.

- 1 Read the five text answers
- 2 Role-play the persona
- 3 Answer 30 BFI-2 items in character
- 4 Reverse-score → aggregate to A C E N O

### QUESTIONNAIRE LIFT (TEST)

**+0.004**

0.3175 → 0.3215, the final push to the top (also +0.005 on Dev).

### Grounded in the literature

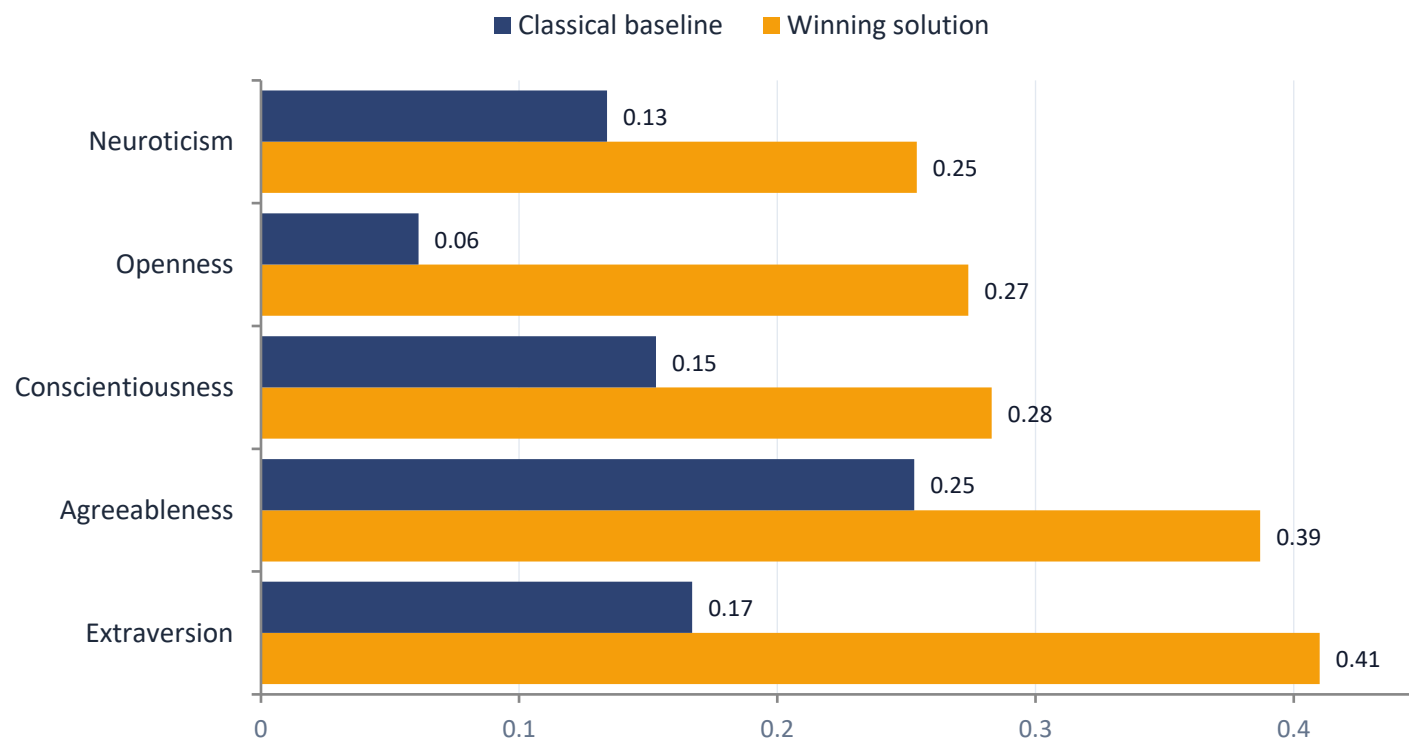
Role-play elicitation (Yang et al. 2024) and persona simulation (Liu et al. 2025) both predict that **answering an instrument in-character** beats direct trait scoring. It did.





## Per-Trait Breakthrough

*Winning solution vs the classical baseline that scored 0.154*



### OPENNESS — THE HARD TRAIT

**0.06 → 0.27**

The trait every prior approach struggled with — more than 4× the baseline.

### Even the floor is high

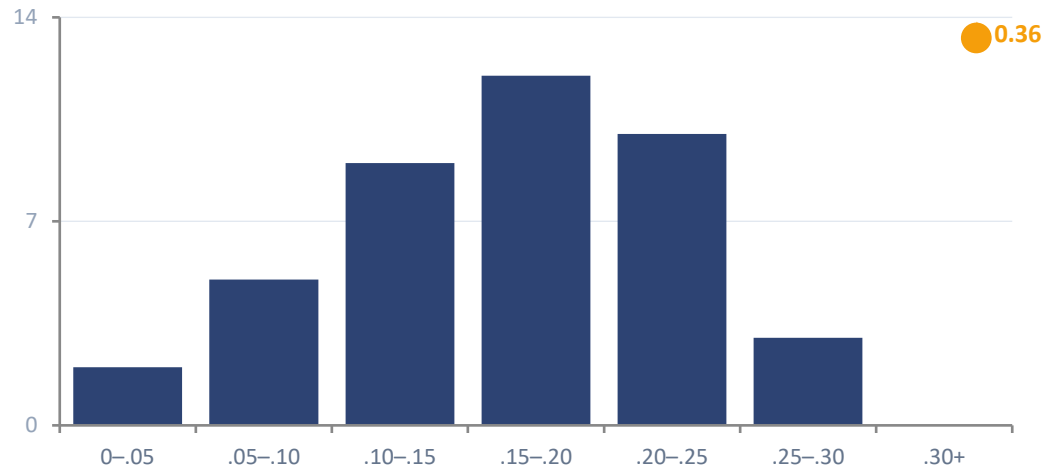
The weakest trait now (Neuroticism, **0.25**) is near the 2019 winner's **overall mean (0.26)**. Extraversion reaches 0.41.



## Both Full Leaderboards

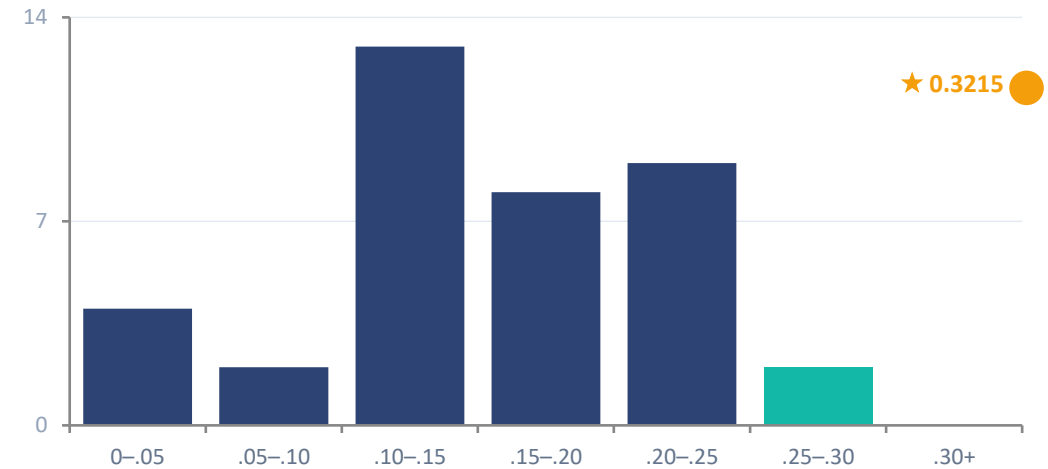
PUBLIC · DEV (BOARD 480)

★ Ours 0.360 — tops the board



PRIVATE · TEST (BOARD 481)

★ Ours 0.3215 — beyond the 2019 tail



Across the entire scored field of 100+ teams, **our entry sits beyond the right edge of both distributions** — above every 2019 submission on the public and private boards alike.



## Built to Generalize



### Leakage-safe protocol

- ✓ Fit only on Train (n=1088); no estimator sees Dev/Test labels
- ✓ Select on Train-CV (out-of-fold) + Dev only
- ✓ Touch the private Test exactly once, via a single frozen entry point
- ✓ Zero-shot LLM outputs use no labels — they cannot leak across splits



### What did NOT work

- ✗ **Plain embedding + Ridge** — lost to TF-IDF (OOF 0.14 vs 0.22)
- ✗ **Bigger MTEB embedders** — bge / gte weaker than e5-large here
- ✗ **Word bigrams / trigrams** — overfit; unigrams generalized better
- ✗ **Cross-trait meta-features** — Train-CV 0.34 but Dev fell to 0.28

# From five short answers to five trait scores.



## The win

0.3215 on the private test — +0.061 over the 2019 winner and above the 2025–26 short-text frontier, verified leakage-free.



## The lever

Zero-shot frontier LLMs — multi-prompt, a second judge, and role-play questionnaire simulation — did what 1,088 rows never could.



## The discipline

Classical bases, honest CV, a single Test touch, and a logged trail of what failed keep the result trustworthy.